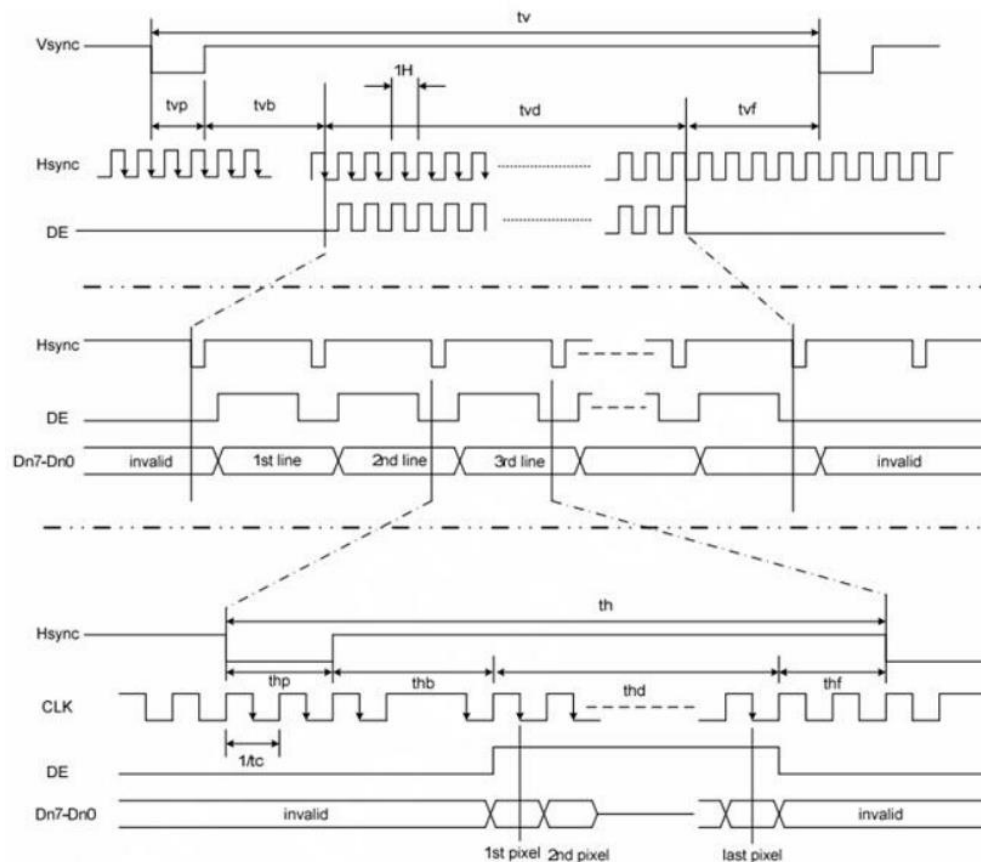
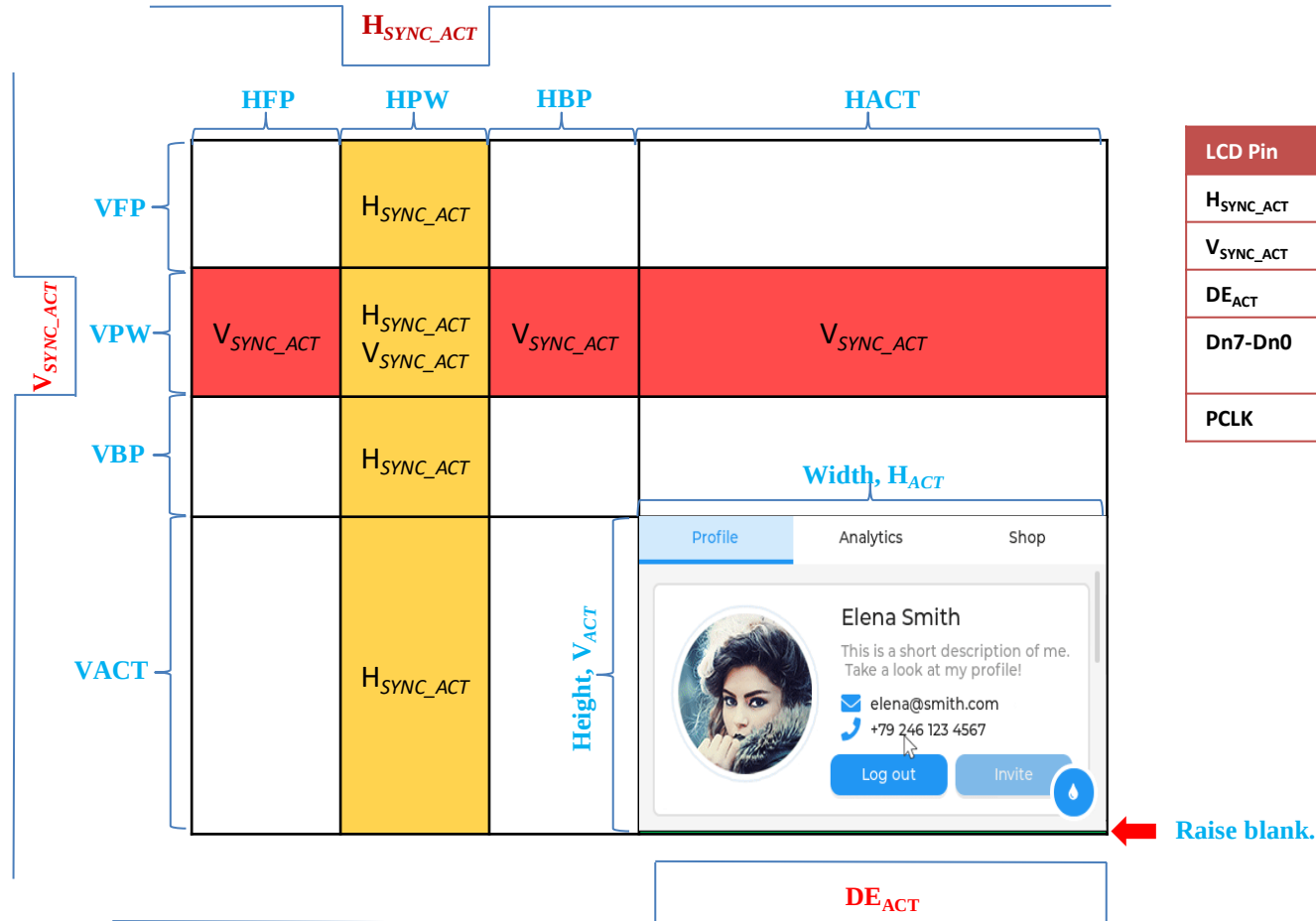


Use EBI-PDMA to drive RGB Sync-type LCM

For NuMicro MCU-family HMI Solution

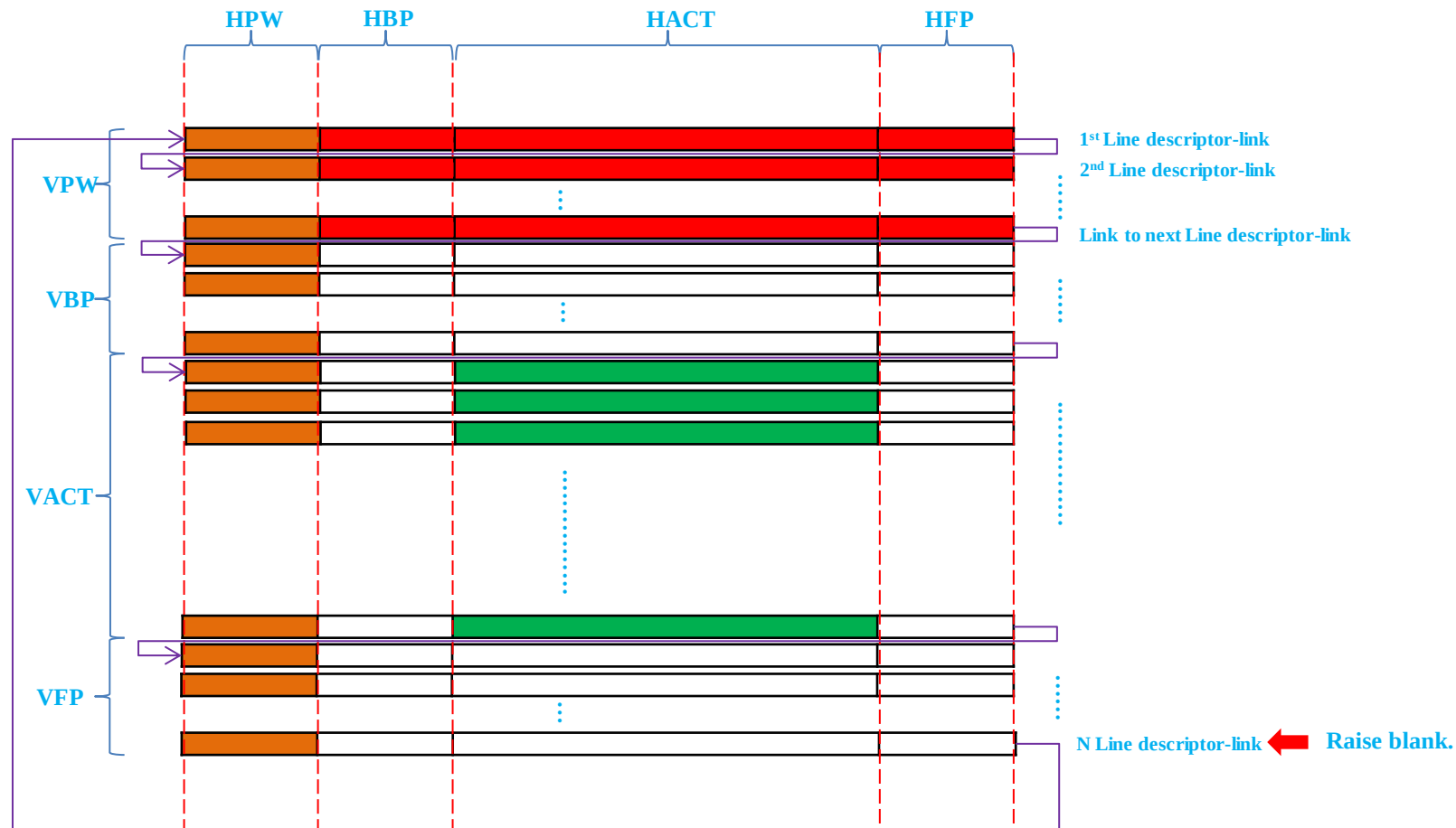
RGB Sync-type LCD Timing

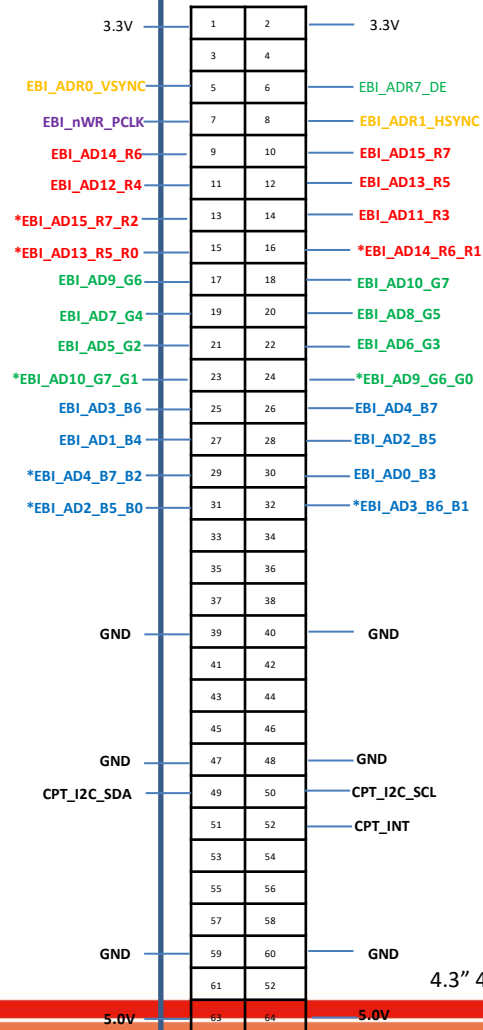
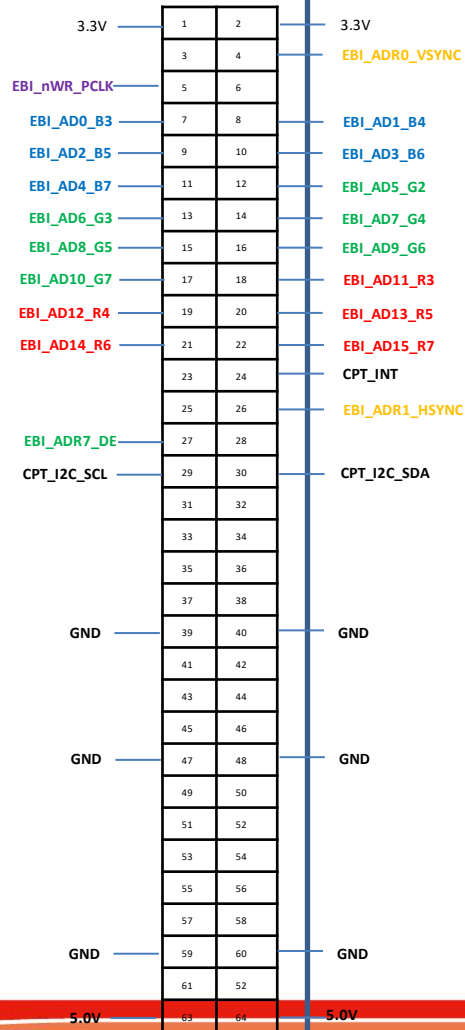




LCD Pin	EBI Pin	EBI Write (16b)
H _{SYNC_ACT}	EBI_ADR1_PH6	Address: 0x00000004
V _{SYNC_ACT}	EBI_ADR0_PH7	Address: 0x00000002
DE _{ACT}	EBI_ADR7_PH0	Address: 0x00000100
Dn7-Dn0	EBI_AD[0-15]	Dummy(Zero) Or RGB565 Pixel data
PCLK	EBI_nWR_PJ9	Every EBI-Writing

Stage	Name
HPW	Horizontal Pulse Width
HBP	Horizontal Back Porch
HACT	Horizontal Active
HFP	Horizontal Front Porch
VPW	Vertical Pulse Width
VBP	Vertical Back Porch
VACT	Vertical Active
VFP	Vertical Front Porch





Driver definitions

```

#define CONFIG_TIMING_HACT      480 /*!< XRES */
#define CONFIG_TIMING_VACT      272 /*!< YRES */
#define CONFIG_TIMING_HBP       30 /*!< HBP (Horizontal Back Porch) */
#define CONFIG_TIMING_HFP       5  /*!< HFP (Horizontal Front Porch) */
#define CONFIG_TIMING_HPWW      41 /*!< HPW (HSYNC plus width) */
#define CONFIG_TIMING_VBP       2  /*!< VBP (Vertical Back Porch) */
#define CONFIG_TIMING_VFP       27 /*!< VFP (Vertical Front Porch) */
#define CONFIG_TIMING_VPW       10 /*!< VPW (VSYNC width) */

#define CONFIG_DISP_VPW_ACTIVE_LOW 1 /*!< Enable VPW active low */
#define CONFIG_DISP_HPWW_ACTIVE_LOW 1 /*!< Enable HPW active low */
#define CONFIG_DISP_DE_ACTIVE_LOW 0 /*!< Disable DE active low */

#define CONFIG_DISP_VSYNC_BITIDX 1 /*!< SET_EBI_ADR0_PH7 */
#define CONFIG_DISP_HSYNC_BITIDX 2 /*!< SET_EBI_ADR1_PH6 */
#define CONFIG_DISP_DE_BITIDX 8 /*!< SET_EBI_ADR7_PH0 */

/* Don't touch me */
#define CONFIG_DISP_VSYNC_BITMASK (1<<CONFIG_DISP_VSYNC_BITIDX)
#define CONFIG_DISP_HSYNC_BITMASK (1<<CONFIG_DISP_HSYNC_BITIDX)
#define CONFIG_DISP_DE_BITMASK (1<<CONFIG_DISP_DE_BITIDX)
#define CONFIG_DISP_VPW_ACTIVE_LOW?(-1*CONFIG_DISP_VSYNC_BITMASK):CONFIG_DISP_VSYNC_BITMASK)
#define CONFIG_DISP_HSYNC_ACTIVE (CONFIG_DISP_HPWW_ACTIVE_LOW?(-1*CONFIG_DISP_HSYNC_BITMASK):CONFIG_DISP_HSYNC_BITMASK)
#define CONFIG_DISP_DE_ACTIVE (CONFIG_DISP_DE_ACTIVE_LOW?(-1*CONFIG_DISP_DE_BITMASK):CONFIG_DISP_DE_BITMASK)
#define CONFIG_DISP_EBI_ADDR (EBI_BANK0_BASE_ADDR+(CONFIG_DISP_EBI*EBI_MAX_SIZE)+\
(CONFIG_DISP_VPW_ACTIVE_LOW<<CONFIG_DISP_VSYNC_BITIDX)+\
(CONFIG_DISP_HPWW_ACTIVE_LOW<<CONFIG_DISP_HSYNC_BITIDX)+\
(CONFIG_DISP_DE_ACTIVE_LOW<<CONFIG_DISP_DE_BITIDX))

```

Driver Enumeration/Structure

```
typedef enum
{
    evHStageHFP,
    evHStageHSYNC,
    evHStageHBP,
    evHStageHACT,
    evHStageCNT
} E_HSTAGE;

typedef enum
{
    evVStageVFP,
    evVStageVSYNC,
    evVStageVBP,
    evVStageVACT,
    evVStageCNT
} E_VSTAGE;

typedef struct
{
    DSCT_T    m_dsch[evHStageCNT];
} S_DSC_HLINE;

typedef struct
{
    #define DEF_TOTAL_VLINES (CONFIG_TIMING_VPW+CONFIG_TIMING_VBP+CONFIG_TIMING_VACT+CONFIG_TIMING_VFP)
    S_DSC_HLINE m_dscV[DEF_TOTAL_VLINES];
} S_DSC_LCD;
```